The documentation and process conversion measures necessary to comply with this revision shall be completed by 20 September 1997

METRIC

MIL-PRF-19500/586C
20 June 1997
SUPERSEDING
MIL-S-19500/586B
22 July 1994

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER,
TYPES 1N5819-1, 1N5819UR-1, 1N6650-1, AND 1N6650UR-1,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, Schottky barrier diodes. Four levels of product assurance are provided for each encapsulated device types as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die.

1.2 Physical dimensions. See figure 1 (DO-41), figure 2 (DO-213AB), figures 3 and 4 (JANC die) dimensions.

1.3 Maximum ratings.

| Types | $V_{RWM}$ 1/ | $I_{O1}$ 2/ | $I_{FSM}$ | $T_{STG}$ | $T_J$ |
|---|---|---|---|---|---|
| | V (pk) | A dc | A (pk) | C° | C° |
| 1N5819-1, 1N5819UR-1 | 45 | 1.0 | 25 | -65°C to +150°C | -65°C to +125°C |
| 1N6650-1, 1N6650UR-1 | 45 | 1.0 | 25 | -65°C to +150°C | -65°C to +150°C |

1/ Derate linearly at 1.8 V/°C above $T_L$ or $T_{EC}$ = +100°C for 1N5819-1, $T_L$ or $T_{EC}$ = +125°C for 1N6650-1 where $T_L$ is at L = .375 inch.

2/ Derate linearly at 14 mA/°C above $T_L$ or $T_{EC}$ = +55°C for 1N5819-1, $T_L$ or $T_{EC}$ = +80°C for 1N6650-1 where $T_L$ is at L = .375 inch.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at $T_A$ = +25°C.

| Types | Max $V_{FM1}$ $I_{FM}$ = 0.1 A | $V_{FM2}$ $I_{FM}$ = 1.0 A | Max $V_{FM3}$ $I_{FM}$ = 3.1 A | Max $I_{RM}$ $V_{RM}$ = 45 V dc pulsed method (see 4.5.1) $T_J$ = +25°C $I_{RM1}$ | $T_J$ = +100°C $I_{RM2}$ | Max $C_T$ $V_R$ = 5 V dc | Max $R_{\theta JL}$ or $R_{\theta JEC}$ 3/8 lead length or end cap | Max $Z_{\theta JX}$ |
|---|---|---|---|---|---|---|---|---|
| | V (pk) | V (pk) | V (pk) | mA | mA | pF | °C/W | °C/W |
| 1N5819-1 | 0.34 | 0.49 | 0.80 | 0.05 | 5.0 | 70 | 70 | 12 |
| 1N5819UR-1 | 0.34 | 0.49 | 0.80 | 0.05 | 5.0 | 70 | 40 | 12 |
| 1N6650-1 | 0.39 | 0.56 | 0.80 | 0.025 | 1.0 | 70 | 70 | 12 |
| 1N6650UR-1 | 0.39 | 0.56 | 0.80 | 0.025 | 1.0 | 70 | 40 | 12 |

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

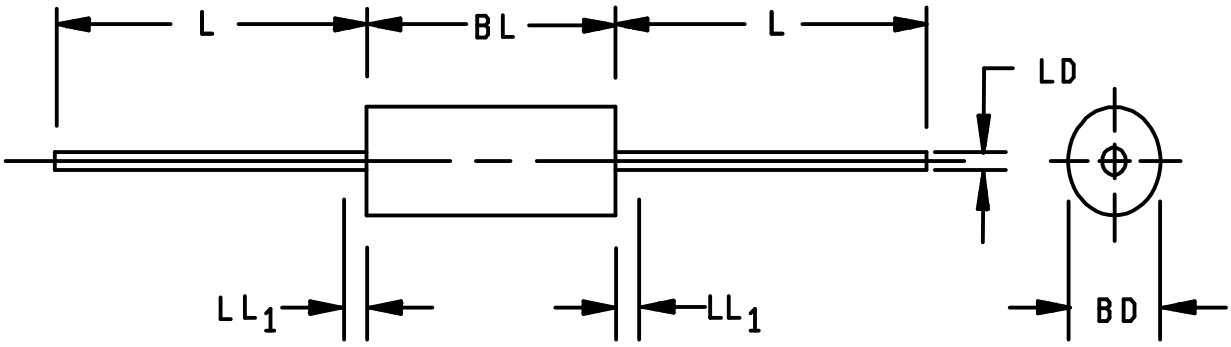
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).



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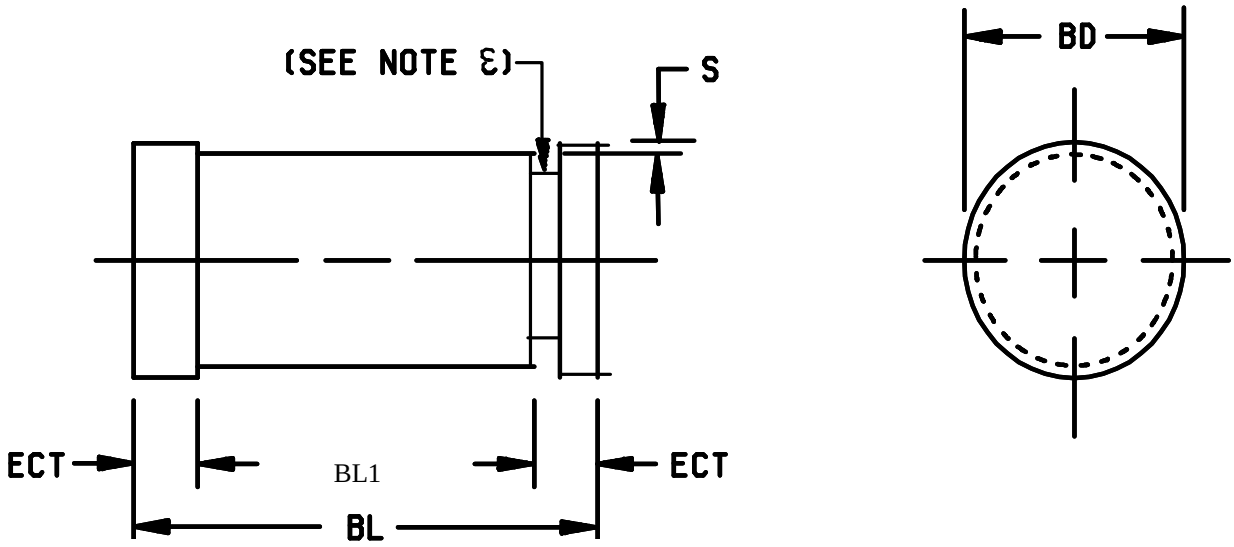

| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Millimeters | | Inches | | |
| | Min | Max | Min | Max | |
| LD | 0.71 | 0.86 | .028 | .034 | |
| BD | 2.03 | 2.72 | .080 | .107 | 3 |
| BL | 4.06 | 5.21 | .160 | .205 | 3 |
| LL | 25.40 | | 1.000 | | |
| $LL_1$ | | 1.27 | | .050 | 4 |

NOTES:
1. Dimensions are in metric.
2. Inch pound equivalents are given for general information only.
3. Package contour optional within cylinder of diameter BD and length BL. Slugs if any shall not be included within this cylinder, but shall not be subject to the minimum limit of BD.
4. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.

FIGURE 1. Physical dimensions, 1N5819-1 and 1N6650-1 (DO-41).



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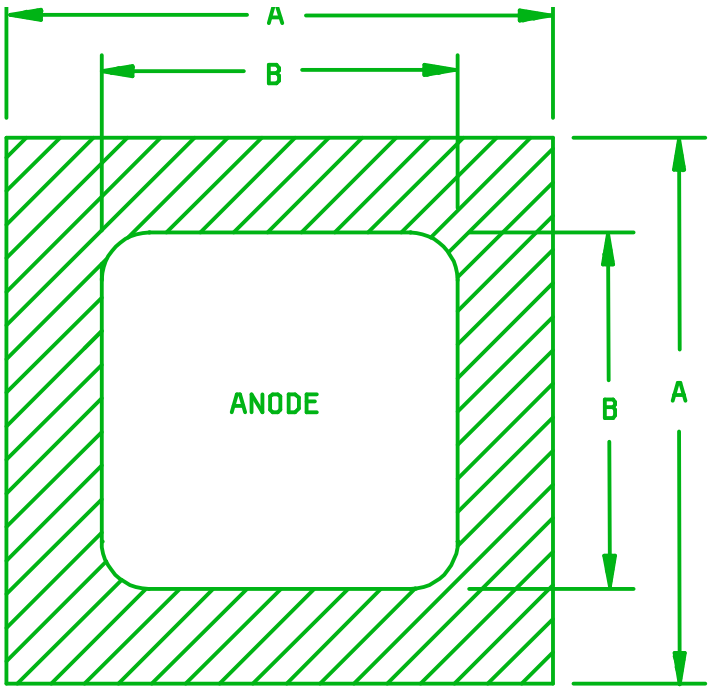

| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Millimeters | | Inches | |
| | Min | Max | Min | Max |
| BL | 4.80 | 5.21 | .189 | .205 |
| $BL_1$ | 4.04 REF | | .159 REF | |
| BD | 2.39 | 2.67 | .094 | .105 |
| ECT | 0.41 | 0.56 | .016 | .022 |
| S | 0.03 | | .001 | |

NOTES:
1. Dimensions are in metric.
2. Inch pounds equivalents are given for general information only.
3. Gap not controlled, shape of body and gap not controlled.

FIGURE 2. Physical dimensions of surface mount family, 1N5819UR-1 and 1N6650UR-1 (DO-213AB).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Millimeters | | Inches | |
| | Min | Max | Min | Max |
| A | 0.89 | 0.99 | .035 | .039 |
| B | 0.79 | 0.84 | .031 | .033 |

Design data

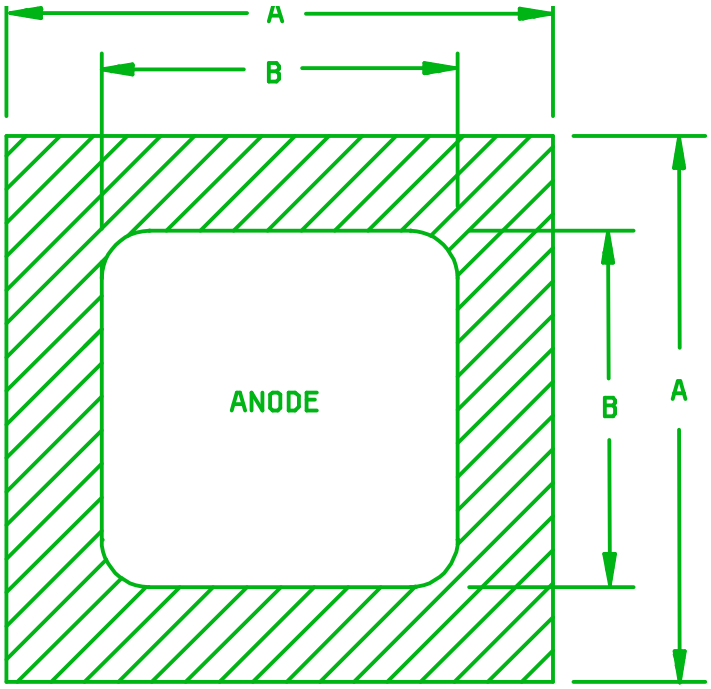
Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness . . . . . 25000 Å min
Gold thickness . . . . . 4000 Å min
Chip thickness . . . . . 10 Mils ± 2 Mils

FIGURE 3. JANC (A-version) die dimensions.



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Millimeters | | Inches | |
| | Min | Max | Min | Max |
| A | 0.94 | 1.04 | .037 | .041 |
| B | 0.84 | 0.89 | .033 | .035 |

Design data

Metallization:
Top: (Anode) . . . . . AL
Back: (Cathode). . . . . Au

AL thickness . . . . . 20000 Å min
Gold thickness . . . . . 5000 Å min
Chip thickness . . . . . 12 Mils ± 2 Mils

FIGURE 4. JANC (B-version) die dimensions.



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3.2 Associated detail specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-19500 and on figures 1, 2, 3 and 4 herein. Plastic packages are prohibited.

3.4.1 Lead material and finish. Lead material shall be copper clad steel with a minimum of 50% copper by weight. Lead finish shall be in accordance with MIL-PRF-19500 and MIL-STD-750. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. These devices shall be constructed in a manner and using materials which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.4.2.1 Surface mount. The U version shall be considered structurally identical to the non-U version except for lead attach.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5.1 Marking for UR devices. UR-suffix parts are to be marked with a cathode band only. Initial container package marking will be in accordance with MIL-PRF-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 JANHC and JANKC devices. Qualification for devices shall be in accordance with MIL-PRF-19500. This testing may be performed on a TO-5 package in lieu of the DO-41 axial leaded package.



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4.3 Screening (JAN, JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with appendix E, table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see Appendix E, table IV of MIL-PRF-19500) | Measurement | | |
|---|---|---|---|
| | JANS level | JANTX and JANTXV levels | JAN level |
| 3a | Temperature cycling | Temperature cycling | Temperature cycling in accordance with MIL-PRF-19500, JANTX level. |
| 3C 1/ | Thermal impedance (see 4.5.4) | Thermal impedance (see 4.5.4) | Thermal impedance (see 4.5.4) |
| 9 | $I_{R1}$ and $V_{FM2}$ | Not applicable | Not applicable |
| 10 2/ | $T_A$ = 110°C; $V_{RWM}$ = 45 V(pk); $I_O$ = 0, half sine wave, f = 60 Hz | $T_A$ = 110°C; $V_{RWM}$ = 45 V(pk); $I_O$ = 0, half sine wave, f = 60 Hz | Not applicable |
| 11 | $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater. $\Delta V_{FM2} \leq \pm$ 50 mV dc. | $I_{R1}$ and $V_{FM2}$ | Not applicable |
| 12 | See 4.3.2 | See 4.3.2, t = 48 hours | Not applicable |
| 13 | Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever greater; $V_{FM2} \leq \pm$ 50 mV dc. | Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial reading or .02 mA whichever is greater; $V_{FM2} \leq \pm$ 50 mV dc. | Not applicable |

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with appendix E, table IV of MIL-PRF-19500, screen 3 prior to this thermal test.

2/ Junction temperature ($T_J$) is not to exceed 115°C with $V_{RWM}$ = 45 V(pk). $T_J$ is affected by the device mounting thermal resistance when parasitic power is generated by the temperature dependent leakage current. Until this leakage becomes significant near thermal runaway, $T_J$ remains approximately equal to $T_A$ or $T_L$ for $I_O$ = 0.

4.3.1 Screening (JANHC or JANKC). Screening of die shall be in accordance with Discrete Semiconductor Die/Chip Lot Acceptance appendix of MIL-PRF-19500 and die shall be 100 percent probed in accordance with group A, subgroup 2.

4.3.2 Burn-in conditions. Burn-in conditions are as follows: $I_F$ = 1.0 A dc (min), $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5);. Mounting and test conditions in accordance with MIL-STD-750, method 1038, test condition B.

4.3.2.1 Mounting. Devices may be mounted using any convenient method including the temporary attachment of leads on UR suffix devices, provided that the parts are burned-in at $T_J \geq$ 110°C +15°, -0°.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-PRF-19500.



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4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. $I_M$ measurement current ..... 1 mA to 10 mA.

b. $I_H$ forward heating current ..... 2.0 A.

c. $t_H$ heating time ..... 10 ms.

d. $t_{MD}$ measurement delay time ..... 70 µs maximum.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 3 | 4066 | $I_{FSM}$ = 25 A(pk), condition A 2, $I_O$ = 1.0 A; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals. |
| 4 | 1036 | $I_F$ = 1.0 A; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. |
| 5 | 1027 | $T_A$ = +75°C ± 25°C; $I_F$ = 1.0 (min) with $I_F$ adjusted as required to achieve an average lot $T_J$ = +125°C +0, -10°C. |
| 6 | 3101 | $R_{\theta JL}$ = 70°C/W; L = .375 inch (9.52 mm) lead length (non-surface mount). $R_{\theta JEC}$ = 40°C/W (surface mount). |

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV of MIL-PRF-19500).

| Subgroup | Method | Condition |
|---|---|---|
| 2 | 4066 | $I_{FSM}$ = 25 A(pk), condition A 2, $I_O$ = 1.0 A; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); 5 surges of 8.3 ms each at 1 minute intervals. |
| 3 | 1027 | $I_O$ = 1.0 A; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; $V_{RWM}$ = 45 V(pk). |
| 4 | 2075 | In accordance with 4.5.2. |

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| 2 | 2036 | Tension: Test condition A; weight = 10 pounds; t = 15 s. Lead fatigue: Test condition E; weight 1 pounds. NOTE: Both tension and lead fatigue are not applicable for US devices. |
| 6 | 1026 | $I_O$ = 1.0 A; $T_A$ = room ambient as defined in the general requirements of MIL-STD-750 (see 4.5); f = 50-60 Hz; $V_{RWM}$ = 45 V(pk). |



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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions for subgroup testing in appendix E, table IX of MIL-PRF-19500 and herein. Electrical measurements (end- points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in (see 4.3.2.1) of MIL-STD-750.

4.5.2 Decap internal visual scribe and break (DPA). Scratch glass at cavity area with diamond scribe. Carefully snap open. Using 30X magnification, examine the area where die or bond material were in contact with the plug and verify metallurgical bonding area.

4.5.3 Steady-state operation life. This test shall be conducted with a half-sine wave of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180° nor less than 150°.

4.5.4 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (appendix E, table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

4.5.4.1 For initial qualification or requalification. Read and record data ($Z_{\theta JX}$) shall be supplied to the qualifying activity on one lot (random sample of 500 devices minimum). Twenty-two serialized devices shall be sent to the qualifying activity for test correlation.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.5.4 | $Z_{\theta JX}$ | | 12 | °C/W |
| Forward voltage | 4011 | $I_{FM}$ = .1 A (pk) pulsed (see 4.5.1) | $V_{FM1}$ | | | |
| 1N5819-1,1N5819UR-1 | | | | | 0.34 | V |
| 1N6650-1,1N6650UR-1 | | | | | 0.39 | V |
| Forward voltage | 4011 | $I_{FM}$ = 1.0 A (pk) pulsed (see 4.5.1) | $V_{FM2}$ | | | |
| 1N5819-1,1N5819UR-1 | | | | | 0.49 | V |
| 1N6650-1,1N6650UR-1 | | | | | 0.56 | V |
| Forward voltage | 4011 | $I_{FM}$ = 3.1 A (pk) pulsed (see 4.5.1) | $V_{FM3}$ | | 0.80 | V |
| Reverse current leakage | 4016 | Pulse method, $V_{RM}$ = 45 V (pk) (see 4.5.1) | $I_{RM1}$ | | | |
| 1N5819-1,1N5819UR-1 | | | | | 0.05 | mA |
| 1N6650-1,1N6650UR-1 | | | | | 0.025 | mA |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +100°C | | | | |
| Reverse current leakage | 4016 | Pulse method (see 4.5.1) | | | | |
| 1N5819-1,1N5819UR-1 | | $V_{RM}$ = 45 V (pk) | $I_{RM2}$ | | 5.0 | mA |
| 1N6650-1,1N6650UR-1 | | $V_{RM}$ = 45 V (pk) | | | 1.0 | mA |
| 1N5819-1,1N5819UR-1 | | $V_{RM}$ = 40 V (pk) | $I_{RM3}$ | | 4.5 | mA |
| 1N6650-1,1N6650UR-1 | | $V_{RM}$ = 40 V (pk) | | | 0.7 | mA |
| 1N5819-1,1N5819UR-1 | | $V_{RM}$ = 35 V (pk) | $I_{RM4}$ | | 3.0 | mA |
| 1N6650-1,1N6650UR-1 | | $V_{RM}$ = 35 V (pk) | | | 0.5 | mA |
| Forward voltage | 4011 | Pulse method (see 4.5.1) $I_F$ = 1.0 A (pk) | $V_{FM4}$ | | | |
| 1N5819-1,1N5819UR-1 | | | | | 0.45 | V |
| 1N6650-1,1N6650UR-1 | | | | | 0.51 | V |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 3 - Continued | | | | | | |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Reverse current leakage | 4016 | Pulse method (see 4.5.1) | $I_{RM5}$ | | | |
| 1N5819-1,1N5819UR-1 | | $V_{RM}$ = 45 V (pk) | | | 0.2 | mA |
| 1N6650-1,1N6650UR-1 | | $V_{RM}$ = 45 V (pk) | | | 0.1 | mA |
| Forward voltage | 4011 | Pulse method (see 4.5.1)<br>$I_F$ = 1.0 A (pk) | $V_{FM5}$ | | 0.65 | V |
| Subgroup 4 | | | | | | |
| Capacitance | 4001 | $V_R$ = 5 V dc, .01 ≤ f ≤ 1 MHz,<br>$V_{SIG}$ = 15 mV p-p | $C_T$ | | 70 | pF |

1/ For sampling plan, see MIL-PRF-19500.



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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

| Inspection | MIL-STD-750 | | Sampling plan |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 22 devices c = 0 |
| Thermal shock (temperature cycling) | 1051 | 500 cycles | |
| Hermetic seal | 1071 | Test condition E | |
| Electrical measurement | | See table I, group A, subgroup 2 | |
| Subgroup 2 | | | 22 devices c = 0 |
| Steady-state reverse bias | 1038 | Test condition A; 1,000 hours, See 4.3, screen 10 | |
| Electrical measurement | | See table I, group A, subgroup 2 | |
| Subgroup 3 | | | |
| Not applicable | | | |
| Subgroup 4 | | | 22 devices c = 0 |
| Thermal resistance, (forward voltage drop diode method) | 4081 or 3101 | $R_{\theta JL}$ = 70°C/W maximum at .375" lead length; $R_{\theta JEC}$ = 40°C/W max.; method 3101 per 4.5.4 except $I_H$ = 500 mA and $T_H$ = 25 sec (min) | |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | |
| Not applicable | | | |

1/ For initial design and process change verification only (one time testing).



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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from : Commander, Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (example JANHCA1N5819) will be identified on the QPL.

| JANC ordering information | | |
|---|---|---|
| | Manufacturer | |
| PIN | 55801 | 59377 |
| 1N5819 | JANHCA1N5819<br>JANKCA1N5819 | JANHCB1N5819<br>JANKCB1N5819 |

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1869)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

**INSTRUCTIONS**

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-PRF-19500/586C | 2. DOCUMENT DATE (YYMMDD)<br>970620 |
|---|---|---|

**3. DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, HERMETIC, DIODE, SILICON, SCHOTTKY BARRIER, TYPES 1N5819-1, 1N5819UR-1, 1N6650-1, AND 1N6650UR-1, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial 614-692-0510<br>(2) AUTOVON 850-0510 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000. | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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